

Multiplicative algorithm for computing D -optimum designs for pVT measurements

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ABSTRACT

Multifactor models are present in industry and in biological experiments; therefore optimum designs are a valuable tool for experimenters, leading to estimators of the parameters with minimum variance. A generalization of the Multiplicative algorithm to find locally D -optimum designs for multifactor models as Tait-like equations is here proposed. The method consists of transforming a conceived set of design points over a finite interval into proportions of the design interval defined by the sub-intervals between successive points. Examples for different modifications of the Tait equation are here presented: a three parameter model under isothermal conditions with pressure as the independent variable and two different modifications of an eight parameter model (Rackett equation) with pressure and temperature as design variables. Optimum design for the first example is equally supported at three points while the two-factor modifications of the Tait equation show designs with more support points than unknown parameters and different weights for each point. Efficiencies of real experimental designs are computed and suitable more efficient experimental designs, satisfying experimental demands are also proposed.

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1. Introduction

Knowledge of the pressure and temperature effects on the volume of dense gases, liquids, solids and liquid mixtures is required in both thermodynamics calculations and engineering processes. The characterization of volume or density as a function of temperature and pressure is particularly important for the design of industrial plants, pipelines and pumps.

The most accurate characterization of density values are crucial for the development of equations of state to calculate phase equilibria and other thermodynamic properties such as enthalpies, entropies, heat capacities and heats of vaporization at given conditions of pressure and temperature. Consequently, the design and optimization of chemical processes are of wide interest.

The Tait equation for pVT data (pressure, volume and temperature) is a suitable empirical equation that conceals accuracy and simplicity [1]. Among other modifications of the equation, the isothermal version is modified in order to include the temperature dependence parameters. Despite the important role played by pVT data, the evaluation of these measurements over wide ranges of pressure and temperature have significant economical and time consuming restrictions.

D -optimum designs, to minimize the generalized variance of the parameter estimates of the Tait equation, are a valuable tool for experimenters. Values of the independent variables leading to estimators of the parameters with minimum variance are a target.

In this case, explicit determination of optimum designs is not a straightforward process. New schemes for developing iterative algorithms, based on the multiplicative algorithm [2], devoted to the numerical construction of optimum designs are proposed in this work. Examples of the application of the proposed algorithms for one-factor and two-factor models are here presented.

1.1. Tait equation

During the experimental investigations of Peter Guthrie Tait, following the scientific expedition of H.M.S. Challenger from 1873 to 1876, a relation between volume and pressure was initially established for fresh water and seawater at different temperatures [1]. Some modifications to the original relation established by Tait were included. The isothermal density for liquids up to elevated pressures is represented by:

$$E(p) = \eta(p) = \frac{\rho_0}{1 - \text{Clog} \frac{B+p}{B+p_0}}, \quad \text{var}(p) = \sigma^2,$$

which is a modification of the original equation proposed by Tammann. Here ρ_0 is the density at a reference pressure p_0 , and B and C are temperature dependent parameters. The selection of p_0 is somewhat arbitrary.

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It should be remarked that the sum $B + p_0$ should be positive and that, therefore, if the parameter B goes negative, p_0 needs to be increased in order to satisfy this constraint [3].

The relation can also be expressed in terms of specific volume instead of density, through the equivalent expression:

$$E(V) = \eta(p) = V_0 \left(1 - C \log \frac{B+p}{B+p_0} \right), \quad \text{var}(V) = \sigma^2 \quad (1)$$

being V_0 the specific volume at the reference pressure p_0 .

The use of the Tait equation in terms of density ρ or specific volume V depends on the experimental equipment and the ability to measure either the density, through vibrating tube densimeters [4], or the volume, by weight using a test sample [5].

These equations are used to obtain the values for B , C and either ρ_0 or V_0 at the reference pressure p_0 . These values are temperature dependent since the Tait equation expressed in the previous forms can only be used to characterize the behavior of density or specific volume under isothermal conditions.

For non isothermal conditions a wide set of modifications appear in literature aiming to simplify and improve the accuracy of the equation to describe the dependence of high-pressure density data for dense gases, liquids and liquid mixtures.

1.2. Modifications of the Tait equation

Temperature dependence of the Tait parameters is usually modeled through linear functions. The Tait B parameter has the strongest temperature dependence and polynomials from degree one to four are used to describe this dependence [3], e.g.

$$B(T) = B_0 + B_1 T + B_2 T^2.$$

The Tait C parameter is a weak function of temperature using either polynomials of degree one or even considering no dependence at all [6].

The dependence of ρ_0 , the density at the reference pressure p_0 , is characterized either by a linear function of the required degree [7,8], e.g.

$$\rho_0(T) = A_0 + A_1 T + A_2 T^2 + A_3 T^3, \quad (2)$$

or by the non-linear function known as the Rackett equation for saturation density [6],

$$\rho_0(T) = A_R / B_R^{[1+(1-(T/C_R))^{D_R}]}. \quad (3)$$

Then, in order to correlate the density values over the temperature and pressure intervals, the following Tait-like equation is used,

$$E(\rho) = \eta(p, T; \theta) = \frac{\rho_0(T)}{1 - C(T) \log \frac{B(T)+p}{B(T)+p_0}}, \quad \text{var}(\rho) = \sigma^2. \quad (4)$$

Pressure, p , and temperature, T , are the two factors or independent variables in this model. We assume that (p, T) can be chosen from the cartesian set $\chi = \mathcal{P} \times \mathcal{T}$, where $\mathcal{P}$ and $\mathcal{T}$ are permissible ranges of values for p and T . Being $\theta^t = (A_0, \dots, B_0, \dots, C_0, \dots)$ the set of unknown parameters.

2. Optimum design background for non-linear models

Consider the general non-linear regression model

$$y = \eta(x; \theta) + \varepsilon, \quad x \in \chi,$$

where the random variables ε are independent and normally distributed with zero mean and constant variance σ^2 and θ is the unknown parameter vector.

Suppose an experiment is to be designed subject to the constraint that the design variable, x , be in a compact subset of a Euclidian space, χ , called design space. Let Ξ be the set of probability distributions on the Borel sets of χ , then any $\xi \in \Xi$ satisfying

$$\int_{\chi} \xi(dx) = 1, \quad \xi(x_i) \geq 0, \quad x \in \chi,$$

is called a design measure, or an approximate design. Kiefer [9], pioneered this approach, and its many advantages are well documented in design monographs, e.g. [10].

The most common method for analyzing data from a non-linear model is based on the use of the linear Taylor series approximation of the response surface [11],

$$\eta(x; \theta) \approx \eta(x; \theta_0) + [\nabla \eta(x; \theta)]_{\theta_0}^t (\theta - \theta_0),$$

where ∇ denotes the gradient with respect to θ and being θ_0 a prior value of θ (for example an estimate calculated on the basis of an initial experiment or suitable values found in the literature).

At this point, it is necessary to assess the degree of non-linearity in the model and to evaluate its effect on the linear approximation. Bates and Watts [12] propose relative measures of intrinsic, I^i , and parameters effect curvature, I^p . Relatively small values for the maximum of both measures indicate that the tangent plane approximation is reasonable [13].

Under these conditions, the variance-covariance matrix of the least square estimator of θ is asymptotically approximated by the inverse of the information matrix induced by ξ , [14],

$$M(\xi, \theta) = \int_{\chi} \nabla \eta(x; \theta) [\nabla \eta(x; \theta)]^t d\xi(x).$$

Let ϕ be a real-valued function defined on the $k \times k$ symmetric matrices and bounded above on $\{M(\xi, \theta) : \xi \in \Xi\}$. The optimum design problem is concerned with finding ξ^* such that $\phi(M(\xi^*, \theta)) = \min_{\xi \in \Xi} \phi(M(\xi, \theta))$ which is called a ϕ -optimum design.

Another important issue in non-linear models is the dependence of the information matrix on the model parameter θ . The dependence of $M(\xi, \theta)$ on the parameters comes from the linearization of the model. Classically, a prior guess of θ_0 for θ is used to design the experiment, with the hope that the local optimum design for θ_0 will be close to the optimum one for the unknown parameters. See, for example, [10] (Chapter 6) and [15]. Along this work locally optimum designs based on prior estimations of the parameters θ_0 will be sought.

Caratheodory's theorem (see, for example, Rockafellar [16], p.p. 153–171) implies that for a model with k parameters, a locally ϕ -optimum design supported at no more than $k(k+1)/2 + 1$ points exists. Therefore, the approximate design approach will be adopted without loss of generality, restricting our attention to designs with a finite set of support points. For convenience, designs will be described using a two row matrix with the different support points displayed in the first row, $x_1, \dots, x_N$, and their corresponding proportion of observations, $\xi(x_i) = \xi_i$, in the second,

$$\xi = \begin{Bmatrix} x_1 & x_2 & \dots & x_N \\ \xi_1 & \xi_2 & \dots & \xi_N \end{Bmatrix},$$

being $\xi_1, \dots, \xi_N$ non negative real numbers which sum up to one.

In this work, the design criterion used for estimating the model parameters is D -optimality. The D -optimum criterion minimizes the volume of the confidence ellipsoid of the parameters and is given by $\phi_D[M(\xi, \theta)] = \det\{M(\xi, \theta)\}^{-1/k}$, where k is the number of parameters in the model. This condition is equivalent to maximize the determinant of the information matrix. It is known that this criterion is a convex and non-increasing function of the designs [10] and so, designs with a small criterion value are desirable. A design that minimizes ϕ_D over all the designs on χ is called a D -optimum design.

An advantage of working with approximate designs is that the optimality of a design can be easily checked. Since the criterion is convex, standard convex analysis arguments using directional derivatives [10], when ϕ is differentiable, show that a design ξ^* is ϕ -optimum if and only if

$$F_\phi[M(\xi^*, \theta), M(\xi_{x_j}, \theta)] \equiv F_\phi(\xi^*, e_j) = d_j - \sum_{j=1}^N \xi_j^* d_j \geq 0, \quad x_j \in \chi, \quad (5)$$

with equality at the support points of ξ^* . Here $M(\xi_{x_j}, \theta)$ is the Information Matrix for a single observation at the point x_j , e_j denotes the j -th unit vector in $\mathbb{R}^N$ and $d_j = \partial \phi / \partial \xi_{x_j}$. The inequality above is referred to as the General Equivalence Theorem [17] in optimum design literature, which provides methods for the construction of optimum designs.

A corollary to the General Equivalence Theorem states that the variance function

$$d(x, \xi, \theta) = [\nabla \eta(x; \theta)]^t M(\xi_D^*, \theta)^{-1} \nabla \eta(x; \theta), \quad (6)$$

evaluated using a ϕ -optimum design achieves its maximum value at the support points of this design. In particular, for the D -optimum criterion this condition proves the optimality of ξ_D^* if and only if $d(x, \xi_D^*, \theta) \leq k$, where k is the number of parameters and the equality is satisfied at least at the support points.

The worth of a design is measured by its efficiency, defined by $\text{eff}_\phi(\xi) = \phi(\xi^*) / \phi(\xi)$. The efficiency can sometimes be multiplied by 100 and reported in percentage. If its value is 50% it means that the design ξ needs to double the total number of observations to perform as good as the optimum design ξ^* .

3. Numerical approximation of optimum designs through multiplicative algorithms to one-factor models

The purpose of experimental design is to select the design points $x_1, \dots, x_N$ and the corresponding weights $\xi_1, \dots, \xi_N$ so that the design ξ is optimum for some criterion, ϕ . Although the number of support points is unknown, according to Caratheodory's theorem, there must exist an optimum design with at most $(k(k+1)/2) + 1$ support points [18].

Consider a model with one design variable x , to be selected from a design interval $\chi = [a, b]$. A simple case is the determination of the best N -point ϕ -optimum design. Let $x_1, x_2, \dots, x_N$ be its support points. Let:

$$W_h = \frac{x_h - x_{h-1}}{b - a}, \quad h = 1, \dots, N+1, \quad (7)$$

where $x_0 = a$ and $x_{N+1} = b$. The variables W_h satisfy

$$W_h \geq 0, \quad \sum_{h=1}^{N+1} W_h = 1.$$

This linear transformation from N variables to $N+1$ variables was proposed by [2].

Thus, we have an example of the following type of optimization problem:

Minimize a criterion $\phi(W, \xi)$ over

$$\mathfrak{P} = \left\{ W = (W_1, \dots, W_{N+1}), \xi = (\xi_1, \dots, \xi_N) : \right. \\ \left. W_h \geq 0, \sum_{h=1}^{N+1} W_h = 1 \text{ and } \xi_j \geq 0, \sum_{j=1}^N \xi_j = 1 \right\}.$$

Then the following simultaneous approaches are used

$$W_h^{(r+1)} = \frac{W_h^{(r)} g(F_h^{(r)}, \delta_1)}{\sum_{t=1}^{N+1} W_t^{(r)} g(F_t^{(r)}, \delta_1)}, \quad \xi_j^{(r+1)} = \frac{\xi_j^{(r)} g(F_j^{(r)}, \delta_2)}{\sum_{i=1}^N \xi_i^{(r)} g(F_i^{(r)}, \delta_2)},$$

where $r=0, 1, \dots$ is the iteration number, $g(F_h, \delta_1)$ and $g(F_j, \delta_2)$, are positive increasing, and if $\delta=0$, $g(F, \delta)$ are constants. Being F_j and F_h the vertex directional derivatives:

$$F_h = F_\phi(W, e_h) = d_h - \sum W_h d_h, \quad d_h = \frac{\partial \phi}{\partial W_h},$$

$$F_j = F_\phi(\xi, e_j) = d_j - \sum \xi_j d_j, \quad d_j = \frac{\partial \phi}{\partial \xi_j}.$$

These iterations neatly ensure the constraints (of non-negativity and summation to 1) of the optimization problem over $\mathfrak{P}$. This type of iteration was first proposed by [19]. As the vertex directional derivatives may take positive and negative values, g needs to be defined over $\mathbb{R}$. On the other hand, due to the Equivalence Theorem, it is desirable that g changes at a reasonable rate in a neighborhood of zero. Several studies about the choice of the function $g(F, \delta)$ can be found in [20], [21], [22], [23]. Through out this work, $g(F, \delta) = \Phi(\delta F)$ will be chosen, where $\Phi(\cdot)$ is the c.d.f. of the standard normal distribution and $\delta > 0$ a suitable constant [2].

To check the optimality of a solution, the first order conditions provide a necessary condition for differential criterion functions ϕ . This condition is necessary and sufficient if the criterion function is also a convex function, like, for example D -optimality. Therefore if ϕ is a convex and differentiable function a design ξ^* will be optimum if and only if,

$$F_h^* = F_\phi(W^*, e_h) \quad \begin{cases} = 0, & \text{for } W_h^* > 0 \\ \geq 0 & \text{for } W_h^* = 0, \end{cases} \quad (8)$$

$$F_j^* = F_\phi(\xi^*, e_j) \quad \begin{cases} = 0, & \text{for } \xi_j^* > 0 \\ \geq 0 & \text{for } \xi_j^* = 0. \end{cases}$$

3.1. Algorithm for obtaining optimum designs: one-factor model

Step 0. Input N , $x_0 = a$, $x_{N+1} = b$, $W^{(0)} = (W_1^{(0)}, \dots, W_{N+1}^{(0)})$, $\xi^{(0)} = (\xi_1^{(0)}, \dots, \xi_N^{(0)})$, δ_1 , δ_2 , *tol*. Set $r=0$.

Step 1. For $j=1, \dots, N$ do,

$$x_j = (b-a)W_j^{(r)} + x_{j-1}, \\ \xi_j^{(r+1)} = \frac{\xi_j^{(r)} g(F_j^{(r)}, \delta_2)}{\sum_{i=1}^N \xi_i^{(r)} g(F_i^{(r)}, \delta_2)}. \quad (9)$$

Step 2. For each $h=1, \dots, N+1$ do,

$$W_h^{(r+1)} = \frac{W_h^{(r)} g(F_h^{(r)}, \delta_1)}{\sum_{t=1}^{N+1} W_t^{(r)} g(F_t^{(r)}, \delta_1)}. \quad (10)$$

Step 3. Stopping rule: If $\min\{F_h^{(r)}, F_j^{(r)}\} \geq -10^{-\text{tol}}$ STOP. Else set $r=r+1$ and return to Step 1.

Under mild conditions which ensure that Eqs. (9, 10) are well-defined, and if ϕ is an increasing and convex function, the algorithm produces a monotonic sequence of values of ϕ [24].

Example 1. To illustrate the use of the algorithm proposed in Section 3.1 the optimum design for Tait Eq. (1) under isothermal conditions has been computed and checked. In the work of [3], the relation between specific volume of chlorodifluoromethane at high pressures and under isothermal conditions has been analyzed. Chlorodifluoromethane has long been used in a variety of air conditioning and refrigeration applications and presents a low ozone depletion potential. Obtaining accurate estimations of the parameters is desirable for the study of their properties and for the consideration of fluid models.

The design space used in this work is the interval $\mathcal{P} = [10, 500]$ MPa and the set of parameters obtained, for constant temperature $T = 298.2$ K, are $\theta_0 = (V_0, B, C)^t = (0.815, 16.81, 0.0982)^t$, with reference pressure $p_0 = 10$ MPa [3]. These values have been considered as best guesses to obtain the locally D -optimum design. A uniform design on $k(k+1)/2 + 1 = 7$ equally spaced points is proposed as an initial design for the algorithm.

The application of the multiplicative algorithm to this non-linear model, with the given nominal values, provides a monotonic sequence of values of the determinant of the information matrix, $\det M(\xi; \theta_0)$, as seen in Fig. 1.

The convergence is achieved and the design,

$$\xi_D^* = \begin{Bmatrix} 10 & 66.5 & 500 \\ 0.33 & 0.33 & 0.33 \end{Bmatrix},$$

with observations at three equally supported points is here obtained. The use of the Equivalence Theorem (6), allows us to check the optimality of ξ_D^* . Eq. (6) is plotted for the whole design space $\mathcal{P}$ (see Fig. 2). It satisfies equality to $k = 3$ at the support points and achieves a local maximum at the only point of the design in the interior of $\mathcal{P}$. The other two support points match the extremes of the design space $p_a = 10$ and $p_b = 500$, achieving Eq. (6) a global maximum.

4. Extension of the multiplicative algorithm to two-factor models

Searching for optimally designed experiments with more than one independent factors is more complicated than for models with a single factor [25]. For the case of pressure–volume relation under non isothermal conditions, variations in either volume or density are explained in terms of pressure and temperature. A wide set of modifications of Tait-like equations are used to characterize this relation, as in Eq. (4). Experiments are then performed for a certain value of pressure and temperature, chosen from the respective design spaces.

$$E[\rho] = \eta(x; \theta), \quad x = (p, T) \in \mathcal{X} = \mathcal{P} \times \mathcal{T} = [p_a, p_b] \times [T_a, T_b].$$

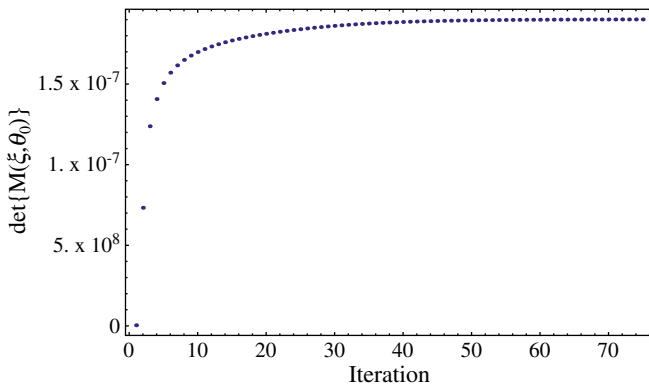


Fig. 1. Values of the $\det\{M(\xi; \theta_0)\}$ for Algorithm 3.1 applied to Example 1.

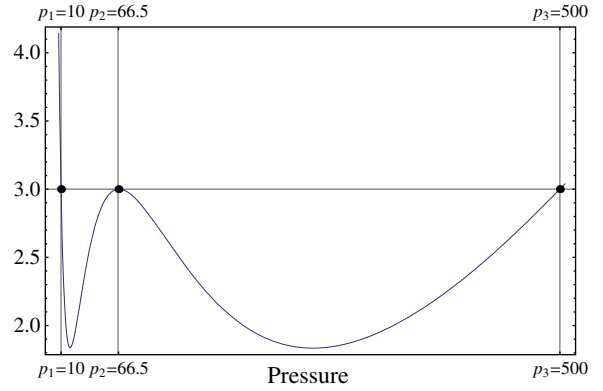


Fig. 2. Fulfillment of the Equivalence Theorem for the D -optimum design proposed in Example 1.

For the extension of the multiplicative algorithm to a two-factor non-linear model, the previous idea based on transforming a set of design points into proportions of the design intervals is used, Eq. (7). The transformation of pressure values, $p_i, i = 1, \dots, N$, into a marginal distribution or set of weights $W_{p_i}, h = 1, \dots, N+1$, is proposed. Then, the corresponding temperatures T paired with each value $p_i, T_{s|p_i}, s = 1, \dots, q$, are transformed into a conditional distribution $W_{T_s|p_i}, s = 1, \dots, q$. This transformation is illustrated in Fig. 3. On the other hand, design weights need to be simultaneously determined. These conditional and marginal distributions must be optimally chosen.

According to Caratheodory's theorem [16,18], we initially assume $k(k+1)/2 + 1$ support points. Thus, the optimization problem can be stated as a problem with respect to $N+2$ distributions. The first is defined by W_p, N are defined by $W_{T_s|p_i}$ (one for each $T_{s|p_i}, i = 1, \dots, N$) and finally one is defined by the design weights, ξ .

This leads to the following simultaneous multiplicative iterations,

$$\begin{aligned} W_{p_h}^{(r+1)} &= \frac{W_{p_h}^{(r)} g(F_{p_h}^{(r)}, \delta_1)}{\sum_{j=1}^{N+1} W_{p_j}^{(r)} g(F_{p_j}^{(r)}, \delta_1)}, h = 1, \dots, N+1, \\ W_{T_s|p_i}^{(r+1)} &= \frac{W_{T_s|p_i}^{(r)} g(F_{T_s|p_i}^{(r)}, \delta_2)}{\sum_{l=1}^{q+1} W_{T_l|p_i}^{(r)} g(F_{T_l|p_i}^{(r)}, \delta_2)}, s = 1, \dots, q+1, i = 1, \dots, N, \\ \xi_t^{(r+1)} &= \frac{\xi_t^{(r)} g(F_t^{(r)}, \delta_3)}{\sum_{i=1}^{qN} \xi_i^{(r)} g(F_i^{(r)}, \delta_3)}, t = 1, \dots, qN, \end{aligned}$$

where $g(F, \delta) = \phi(\delta F)$ as in Section 3, and $F_p, F_{T|p}, F_t$ are the vertex directional derivatives. There are necessary and sufficient optimality conditions equivalent to those in Eq. (8) if ϕ is convex and differential on its feasible region.

4.1. Algorithm for obtaining optimum designs: two-factor model

Step 0. Input $N, q, p_0 = p_a, p_{N+1} = p_b, T_{0|p} = T_a, T_{q+1|p} = T_b, W_p^{(0)}, W_{T|p_i}^{(0)}, \dots, W_{T|p_i}^{(0)}, \xi^{(0)}, \delta_1, \delta_2, \delta_3, \text{tol}$. Set $r = 0$.

Step 1. At iteration r the qN design points are supported at:

$$\begin{aligned} &\text{For } j = 1, \dots, N, \text{ do,} \\ &\quad \bullet p_j = (p_b - p_a)W_{p_j}^{(r)} + p_{j-1}. \\ &\quad \bullet \text{For } s = 1, \dots, q, \text{ do,} \\ &\quad \quad T_{s|p_j} = (T_b - T_a)W_{T_{s|p_j}}^{(0)} + T_{s-1|p_j}. \end{aligned}$$

Taking $\xi_t^{(r)}, t = 1, \dots, qN$ proportion of observations at each point.

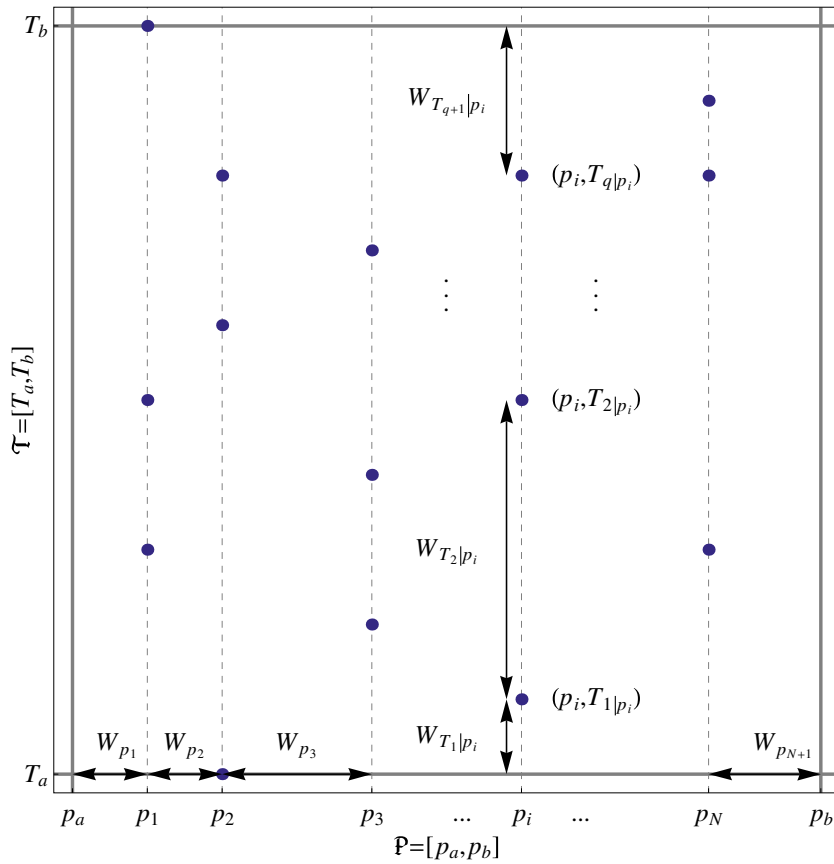


Fig. 3. Transformation of the design space $\chi = \mathcal{P} \times \mathcal{T}$ into proportions of the design intervals, W_h and $W_{T_i|p}$.

Step 2. Update the marginal distribution of pressure, p .

For $h = 1, \dots, N+1$, do,

$$\bullet W_{p_h}^{(r+1)} = \frac{W_{p_h}^{(r)} \phi(\delta_1 F_{p_h}^{(r)})}{\sum_{j=1}^{N+1} W_{p_j}^{(r)} \phi(\delta_1 F_{p_j}^{(r)})}.$$

Step 3. Update the conditional distributions of temperature for each value of variable p .

For $i = 1, \dots, N$, do,

• For $s = 1, \dots, q+1$, do,

$$W_{T_s|p_i}^{(r+1)} = \frac{W_{T_s|p_i}^{(r)} \phi(\delta_2 F_{T_s|p_i}^{(r)})}{\sum_{l=1}^{q+1} W_{T_l|p_i}^{(r)} \phi(\delta_2 F_{T_l|p_i}^{(r)})}.$$

Step 4. Update the set of weights of the design.

For $t = 1, \dots, Nq$, do,

$$\bullet \xi_t^{(r+1)} = \frac{\xi_t^{(r)} \phi(\delta_3 F_t^{(r)})}{\sum_{i=1}^{qN} \xi_i^{(r)} \phi(\delta_3 F_i^{(r)})}.$$

Step 5. Stopping rule: If $\min\{F_p^{(r)}, F_{T|p}^{(r)}, F_t^{(r)}\} \geq -10^{-tol}$ STOP. Else $r = r+1$, and return to Step 1.

In order to illustrate the usefulness of the algorithm, two examples from real experiments of pVT measurements are solved.

Example 2. The algorithm proposed in Section 4.1 has been used to obtain the optimum experimental design for the modified Tait Eq. (4). The modified Tait equation is a non-linear model with pressure and temperature as independent variables. In the work of [7] the densities of 1-phenylundecane have been measured up to 65 MPa, between 293.15 K and 353.15 K. To correlate the measured values over the temperature and pressure intervals, Eq. (4) has been used, where $\rho_0(T)$ is the temperature dependence of the density at pressure $p_0 = 0.1$ MPa and is expressed as a polynomial function of degree up to 3, see Eq. (2). The dependence of parameter $B(T)$ is given by a quadratic polynomial function and C was assumed to be temperature independent. Obtaining adequate estimations of the Tait parameters allows the estimation of the isobaric thermal expansivity and the isothermal compressibility. Its dynamic characterization is important in petroleum engineering.

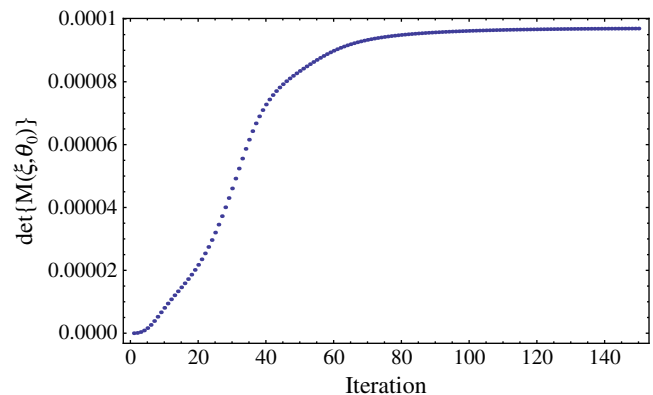


Fig. 4. Evolution of $\det\{M(\xi, \theta_0)\}$ for Algorithm 4.1 applied to Example 2.

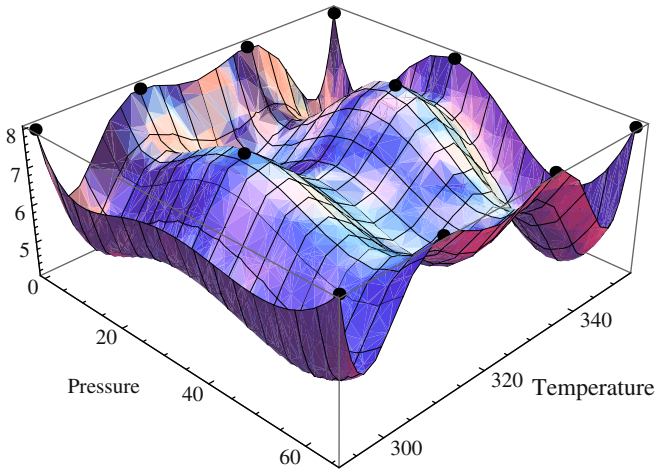


Fig. 5. Fulfillment of the Equivalence Theorem for the D -optimum design proposed in Example 2.

The design space used is $X = P \times T = [0.1, 65] \text{ MPa} \times [293.15, 353, 15] \text{ K}$ and the set of best guesses of the parameters are obtained from [7], $\theta_0 = (A_0, A_1, A_2, A_3, B_0, B_1, B_2, C)^t = (0.97877, -7.8964 \cdot 10^{-7}, -1.9601 \cdot 10^{-6}, 1.8159 \cdot 10^{-9}, 480.83, -1.6932, 0.0016652, 0.087944)^t$, $p_0 = 0.1 \text{ MPa}$. These values are necessary to obtain the locally D -optimum design.

Algorithm 4.1 produces a monotonic sequence of values of $\det M(\xi; \theta_0)$, and convergence is reached, as shown in Fig. 4. The obtained design, ξ_D^* is supported at 11 points with different proportions of observations for each point. In this case, 8 of the support points lie at the boundaries of the design space χ , while two of them belong to its interior.

$$\xi_D^* = \left\{ \begin{array}{cc} (0.1, 293.15) & 0.12 \\ (0.1, 311.99) & 0.07 \\ (0.1, 333.55) & 0.10 \\ (0.1, 353.15) & 0.12 \\ (28.58, 308.91) & 0.05 \\ (28.58, 339.50) & 0.07 \\ (28.58, 353.15) & 0.07 \\ (65, 293.15) & 0.12 \\ (65, 311.61) & 0.07 \\ (65, 334.53) & 0.09 \\ (65, 353.15) & 0.12 \end{array} \right\}^t. \quad (11)$$

The use of the Equivalence Theorem allows us to check D -optimality of the design ξ_D^* . Eq. (6) is plotted over the design space χ (see Fig. 5). We see that equality to $k=8$ is satisfied at the support points, being 8 the number of unknown parameters in the model.

Example 3. Measurements of pVT data of fuels and rocket propellants are of interest to develop physical property models that will aid in the optimum use of these fluids and efficient engine design. In the work of [26], measurements are taken over JP-10, which is the current fuel used in the United States for air-breathing missiles. Because of its high thermal stability, high energy density, low cost, and widespread availability, JP-10 is being investigated as a fuel to be used in pulse-detonation engines.

Measurements are taken over $\chi = P \times T = [0.083, 30] \text{ MPa} \times [270, 470] \text{ K}$. Density data for the compressed liquid were correlated with the Tait equation, Eq. (4), where $\rho_0(T)$ is expressed by using the Rackett equation, Eq. (3). The parameter $B(T)$ was expressed by a quadratic polynomial, and the temperature dependence of C was omitted. The set of estimated parameters in [26] have been used as initial best guesses for obtaining the D -optimum design, $\theta_0 = (A_R, B_R,$

$C_R, D_R, B_0, B_1, B_2, C)^t = (293.822, 0.504584, 621.385, 0.574121, 417.1, -1.35603, 1.15586 \cdot 10^3, 79.1823 \cdot 10^3)^t$, being $p_0 = 0.083 \text{ MPa}$ the reference pressure.

As in the previous example Algorithm 4.1 produces a monotonic sequence of values of $\det M(\xi; \theta_0)$. As result of the convergence of the algorithm, a D -optimum design is obtained. ξ_D^* is supported at 10 points with only one of them in the interior of χ .

$$\xi_D^* = \left\{ \begin{array}{cc} (0.083, 270) & 0.11 \\ (0.083, 334.95) & 0.08 \\ (0.083, 421.72) & 0.12 \\ (0.083, 470) & 0.12 \\ (12.51, 439.1) & 0.02 \\ (12.60, 470) & 0.12 \\ (30, 270) & 0.11 \\ (30, 337.94) & 0.09 \\ (30, 425.91) & 0.11 \\ (30, 470) & 0.12 \end{array} \right\}^t.$$

The Equivalence Theorem proves D -optimality of the design. The plot of Eq. (6) over the design space χ , Fig. 6, shows the attainment of equality to $k=8$ at the support points.

5. Discussion

The reliable knowledge of the pVT behavior of pure compounds and mixtures is of great importance in many fields and in industrial practice. Accurate estimations of the equation parameters are a key target.

Optimum designs for non-linear models like the Tait equations are based on the use of a linear Taylor approximation. The models used in Examples 1, 2 and 3 have been fitted to pVT data that can be found in [3,7,26]. Measures of curvature for non-linear models quantify the deviation of the model from linearity. To ensure that the linear approximation is satisfactory, the values of these curvatures Γ^q and Γ^θ should be lower than $1/\sqrt{F_{k,N-k;1-\alpha}}$, where $F_{k,N-k;1-\alpha}$ is the $100(1-\alpha)$ percentile of the F distribution with $(k, N-k)$ degrees of freedom [12,13,14]. Algorithms to obtain these curvature measures are implemented in statistical packages as for example R, SAS or Mathematica. For the three examples in our work, curvature measurements take values lower than $1/\sqrt{F_{k,N-k;0.95}}$, and the linear approximation is appropriate. For non-linear models with strong curvature D -optimality

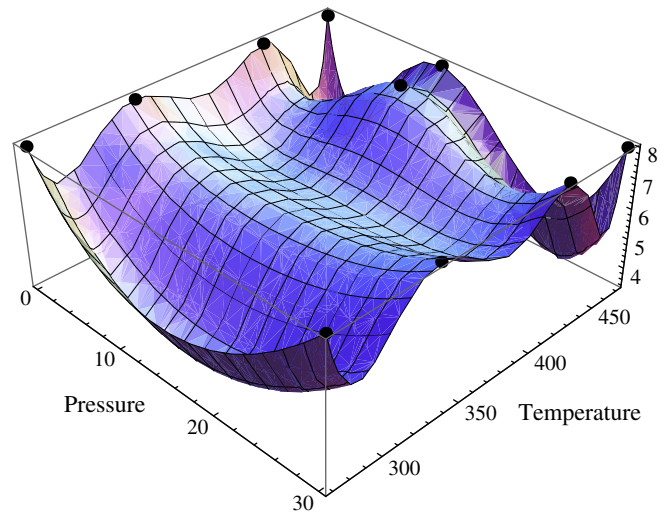


Fig. 6. Fulfillment of the Equivalence Theorem for the D -optimum design proposed in Example 3.

Table 1
Efficiencies of uniform distributed designs over $\mathcal{P}$ for Example 1. $\#p$ are the number of support points of the compared uniform designs. Different columns compare experimental designs with k times more observations at the support points at the boundaries of $\mathcal{P}$ than in the points of the interior.

# p\k	1	2	3	4	5
3	63.45	60.43	56.56	53.29	50.60
4	70.34	71.48	69.22	66.58	64.08
5	71.45	75.47	74.84	73.10	71.17
6	70.68	76.67	77.39	76.53	75.19
7	69.26	76.60	78.40	78.30	77.45
8	67.65	75.92	78.57	79.13	78.78
9	66.02	74.96	78.27	79.38	79.47
10	64.46	73.86	77.71	79.28	79.75

criterion can be replaced by other criterions as Q-optimality that uses a quadratic approximation [27].

Optimum designs are a valuable tool for measuring the worth of an experimental design, ξ_{exp} , through the efficiency, $eff_{\phi_D}(\xi_{exp}) = \phi_D(\xi_D^*)/\phi_D(\xi_{exp})$. This efficiency is usually reported in percentage. The efficiency is the percentage of observations which the optimum design would need to achieve the accuracy of the experimental design being compared.

Optimum designs are usually criticized for taking observations at extreme points in their design spaces. Our examples show that only one or two support points lie in the interior of χ while the designs used in practice are usually supported on more distinct points than the optima. In Example 1, [3] performs observations over 23 different values of pressure, almost uniformly distributed. The experimental design of this work achieves an efficiency of 57.50% in comparison with the optimum design obtained. The work of [7] takes 14×7 different observations with different values of pressure and temperature uniformly distributed over the design space. In this case the efficiency of the experimental design reaches 62.37%. Finally, [26] takes single measurements at 12×11 , a design with an efficiency of 55.34%.

Due to the use of uniform patterns in the distribution of the observations over the design space, the efficiencies of uniform designs with different levels of pressure and temperature have been calculated. The first column of Table 1 presents the efficiencies for uniform designs from 3 to 10 different pressure levels, uniformly distributed over $\mathcal{P}$, with $1/N$ observations to be taken at each point.

Tables 2 and 3 present the efficiencies of experimental designs with different levels of pressure and temperature, with their points uniformly distributed over $\mathcal{P} \times \mathcal{T}$. All of these experimental designs have been supposed to be equally weighted with $1/N$ observations at each support point.

As stated before, D -optimum designs usually take extreme observations. A way of increasing the efficiency of uniformly distributed experimental designs is by increasing the weight of the experimental

Table 2
Efficiencies of uniform distributed designs over $\mathcal{P} \times \mathcal{T}$ for Example 2. $\#p$ are the number of different values of pressure and $\#T$ the values of temperature. The uniform design would be supported at $\#p \times \#T$ different points.

# p\# T	4	5	6	7	8	9	10
4	86.58	83.17	79.67	76.86	74.64	72.86	71.42
5	81.93	78.70	75.39	72.73	70.62	68.94	67.58
6	78.82	75.72	72.54	69.97	67.95	66.33	65.02
7	76.62	73.61	70.51	68.02	66.05	64.48	63.21
8	74.99	72.04	69.01	66.57	64.64	63.11	61.86
9	73.73	70.83	67.85	65.45	63.56	62.05	60.82
10	72.73	69.87	66.93	64.56	62.69	61.20	59.99

Table 3
Efficiencies of uniform distributed designs over $\mathcal{P} \times \mathcal{T}$ for Example 3. The uniform design would be supported at $\#p \times \#T$ different points.

# p\# T	4	5	6	7	8	9	10
4	78.20	76.39	73.43	70.79	68.61	66.82	65.35
5	74.06	72.34	69.54	67.03	64.96	63.27	61.87
6	71.28	69.63	66.93	64.52	62.52	60.89	59.55
7	69.31	67.70	65.07	62.73	60.79	59.21	57.90
8	67.84	66.27	63.70	61.40	59.51	57.95	56.68
9	66.70	65.16	62.63	60.38	58.51	56.99	55.73
10	65.80	64.28	61.78	59.56	57.72	56.22	54.98

points that lie at the boundary of $\mathcal{P}$. Columns from two to five of Table 1 present the efficiencies of uniform distributed designs with k times more observations in the extreme points of $\mathcal{P}$, p_a and p_b , than on the points in the interior.

Tables 4 and 5 present the efficiencies of experimental designs with the same number of levels for pressure as for temperature, $\#p = \#T$. These designs are uniformly distributed over $\mathcal{P} \times \mathcal{T}$ with k times more observations in the $2(\#p + \#T - 2)$ experimental points at the boundary of the design space than on the points in the interior.

Optimal designs will actually depend on the initial best guesses, θ_0 , of the parameters and will vary with different values of these guesses. These locally optimum designs are of interest because they provide a useful reference point for other designs and they may be stable over a range of θ values when choosing an adequate nominal value [11].

Initial best guesses can sometimes be found in literature, for example, the Dortmund Data Bank continuously updates and extends thermodynamic and thermophysical data [28]. Therefore, optimum designs which would provide better estimations with lower experimental costs are of wide interest.

In our examples the initial best guesses come from real experimental works, [3,7,26], and the optimum designs obtained are robust against deviations of the initial nominal values of the parameters. To assess the effect of variations in the prior estimates of parameters, optimum designs were obtained using a variety of initial estimates of the parameters. The prior estimates were generated by changing each parameter $\pm 20\%$ from the values provided in [7]. Efficiencies of the obtained design (Eq. (11)) relative to the 2^8 optimum designs obtained from the variations of the 8 prior parameters of Example 2 have been calculated. For all feasible cases the efficiency was never lower than 77% and greater than 95% for most of the variations. This shows that the D -optimum design (Eq. (11)) is not very sensitive to changes in the initial estimates θ_0 .

To achieve some robustness with respect to a misestimation of θ several approaches have been suggested [18,29,30]. If a prior distribution of the parameters is available, the Bayesian design maximizes the expectation of the criterion function. In minimax optimum design the criterion function is replaced by its worst possible value for θ .

Table 4
Efficiencies of uniform distributed designs over $\mathcal{P} \times \mathcal{T}$ for Example 2 with $\#p = \#T$ and $\#p \times \#T$ different experimental points. The designs have k times more weight on the points at the boundary of the design space than in the points of the interior.

# p=# T\k	1	2	3	4	5
4	86.58	89.46	89.86	89.84	89.74
5	78.70	84.30	85.74	86.20	86.34
6	72.54	79.89	82.20	83.12	83.53
7	68.02	76.45	79.46	80.80	81.47
8	64.64	73.71	77.26	78.96	79.87
9	62.05	71.47	75.44	77.44	78.57
10	59.99	69.58	73.87	76.14	77.46

Table 5

Efficiencies of uniform distributed designs over $\mathcal{P} \times \mathcal{T}$ for Example 3 with $\#p = \#T$ and $\#p \times \#T$ different experimental points. The designs have k times more weight on the points at the boundary of the design space than in the points of the interior.

$\#p = \#T \backslash k$	1	2	3	4	5
4	78.20	81.69	82.63	83.01	83.20
5	72.34	78.13	79.95	80.78	81.22
6	66.93	74.16	76.72	77.93	78.60
7	62.73	70.85	73.99	75.54	76.42
8	59.51	68.15	71.73	73.58	74.67
9	56.99	65.92	69.85	71.95	73.21
10	54.98	64.04	68.23	70.55	71.97

Sequential designs give in each step more and more information about the initial parameter choice. These approaches present significant difficulties in the case of multifactor non-linear models and further work needs to be done.

Although most of the optimum experimental design theory is developed for one-factor models, models with several independent variables, such as the Tait equation, are frequently used in experimental sciences. New methods are therefore needed to solve the optimization problem of finding optimum designs.

The multiplicative algorithm, initially presented by [31] and [20], and its different modifications, as well as its applications to different types of problems, show the importance of this first order algorithm in the search of optimum designs ([2], [23], [24], [32], [33] and [34]). Due to its properties and its ease of implementation criteria, this family of algorithms is attracting the attention of researchers, as shown in the mentioned set of recent works.

Finally, the new proposal presented in this work, shows the effectiveness and adequacy of these methods for obtaining D -optimum designs for two-factor models. A further generalization of this method to applications with more than two-factors is currently being developed.

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